

Q16

iii

$$\left(\frac{n^4}{12} + \frac{12}{n^4} \right) \left(\frac{n^8}{144} + \frac{144}{n^8} + 1 \right)$$

$$\frac{a}{9} + \frac{b}{6} \quad (a^2 + ab + b^2) = a^3 + b^3$$

$$\left(\frac{n^4}{12} + \frac{12}{n^4} \right) \left(\frac{n^4}{11} \right)^2 + \left(\frac{n^4}{12} \right) \left(\frac{12}{n^4} \right) + \left(\frac{12}{n^4} \right)^2$$

$$\left(\frac{n^4}{12} + \frac{12}{n^4} \right) \left(\frac{n^4}{11} \right)^2 + \left(\frac{12}{n^4} \right)^2 = \left(\frac{n^4}{12} \right)^3 + \left(\frac{12}{n^4} \right)^3$$

$$= \frac{n^{12}}{1728} - \frac{1728}{n^{12}}$$

=> iv

$$\left(c^2 - \frac{1}{c^2} \right) \left(\frac{c^4}{c^4} + \frac{1}{c^4} + 1 \right)$$

so ↓ ↓
a - b

we when take $(a^3 - b^3) = (a-b)(a^2 + ab + b^2)$

$$\left(c^2 - \frac{1}{c^2} \right) \left(c^4 + 1 + \frac{1}{c^4} \right) =$$

$$c^2 - \frac{1}{c^2} \quad c^4 + 1 + \frac{1}{c^4} = ()^3 - ()^3$$

$$= (c^2)^3 - \left(\frac{1}{c^2} \right)^3$$

$$= c^6 - \frac{1}{c^6} \text{ Ans}$$